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Color Transfer

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Abstract

Color transfer is the process of adjusting the colors of a source image to match the style or color distribution of a target image. This report presents a comprehensive study of three color transfer methods: (1) per-channel histogram matching, (2) mean and covariance matching via an affine transform, and (3) Sliced Wasserstein Projection (SWP), an iterative gradient-descent-based optimal transport approach.

We describe the theory behind each method – from marginal histogram alignment, to global first- and second-order statistical matching using covariance whitening and re-coloring, to the concept of sliced Wasserstein distance that motivates SWP. Implementation details are presented, including our transition from NumPy prototypes to a significantly faster PyTorch implementation for the SWP method. The SWP pipeline includes sampling multiple random 1D projections per iteration, applying a step-size parameter τ (empirically tuned to 5.4), and using a dual-epsilon convergence criterion to ensure stable and efficient stopping.

After that we show examples of the artifacts that might emerge from color transfer, and we show how to correct these artifacts, using Transportation Map Regularization (TMR). We then present the limitations of color transfer.

1 Introduction

Recoloring an image to adopt the "look and feel" of another is an important problem in image editing and computer vision, commonly referred to as *color transfer*. Given a **target image** that defines a desired color style and a **source image** (with different original colors), the goal is to transform the source image's colors so that the result appears as if it were photographed or painted with the color palette of the target image. Applications of color transfer range from subtle photo enhancements to dramatic scene alterations (for example, making a daytime scene appear as if it were shot at sunset or matching the color grading between movie shots).
In this project, we explore three representative approaches:

- **Histogram Matching (per channel):** A simple method that independently remaps each color channel of the source to match the cumulative histogram of the source.
- **Mean and Covariance Matching:** A linear color transfer method that aligns the source's mean vector and covariance matrix with those of the target.
- **Sliced Wasserstein Projection (SWP):** An advanced iterative technique based on the sliced Wasserstein distance, which repeatedly matches the source and target distributions along random 1D directions.

The remainder of this report presents the three color transfer methods and their implementations, analyzes their results, discusses typical artifacts, their correction using TMR and the limitations of these methods, and ends with a conclusion, summarizing the main findings.

2 Histogram Matching (Per-Channel)

Histogram matching is a well-known image processing technique. In this method, each color channel is processed independently. The basic idea can be understood through a simple sorting-based procedure: for a given channel $c \in \{R, G, B\}$, the source intensities and the target intensities are both sorted, and the smallest value in the source is assigned the smallest value in the target, the second smallest is assigned the second smallest, and so on. In probabilistic terms, this process corresponds to matching the empirical cumulative distribution functions (CDFs). If $F_c^{(s)}$ and $F_c^{(t)}$ denote the empirical CDFs of the source and target channels, respectively, then each source intensity x is transformed using

$$T_c(x) = \left(F_c^{(t)}\right)^{-1}\left(F_c^{(s)}(x)\right),$$

meaning that a pixel in the source channel is mapped to the intensity in the target channel that has the same percentile. Repeating this operation independently for the R, G, and B channels produces an output image whose marginal color distributions match those of the target.

Although straightforward and computationally inexpensive, this method ignores correlations between color channels. Because the channels are matched independently, the resulting RGB triplets may not reflect natural color relationships, sometimes leading to unrealistic hues or distortions. As a result, per-channel histogram matching is effective for transferring global tonal characteristics but generally insufficient for achieving coherent or perceptually consistent color transfer.

The first experiment was performed by applying per-channel histogram matching directly in the RGB color space. The resulting image, shown in Figure 1, exhibits poor results, and the overall style of the target image is not reproduced faithfully. This poor performance is expected, as the RGB channels are highly correlated in natural images, and modifying each channel independently breaks the geometric relationships between them.

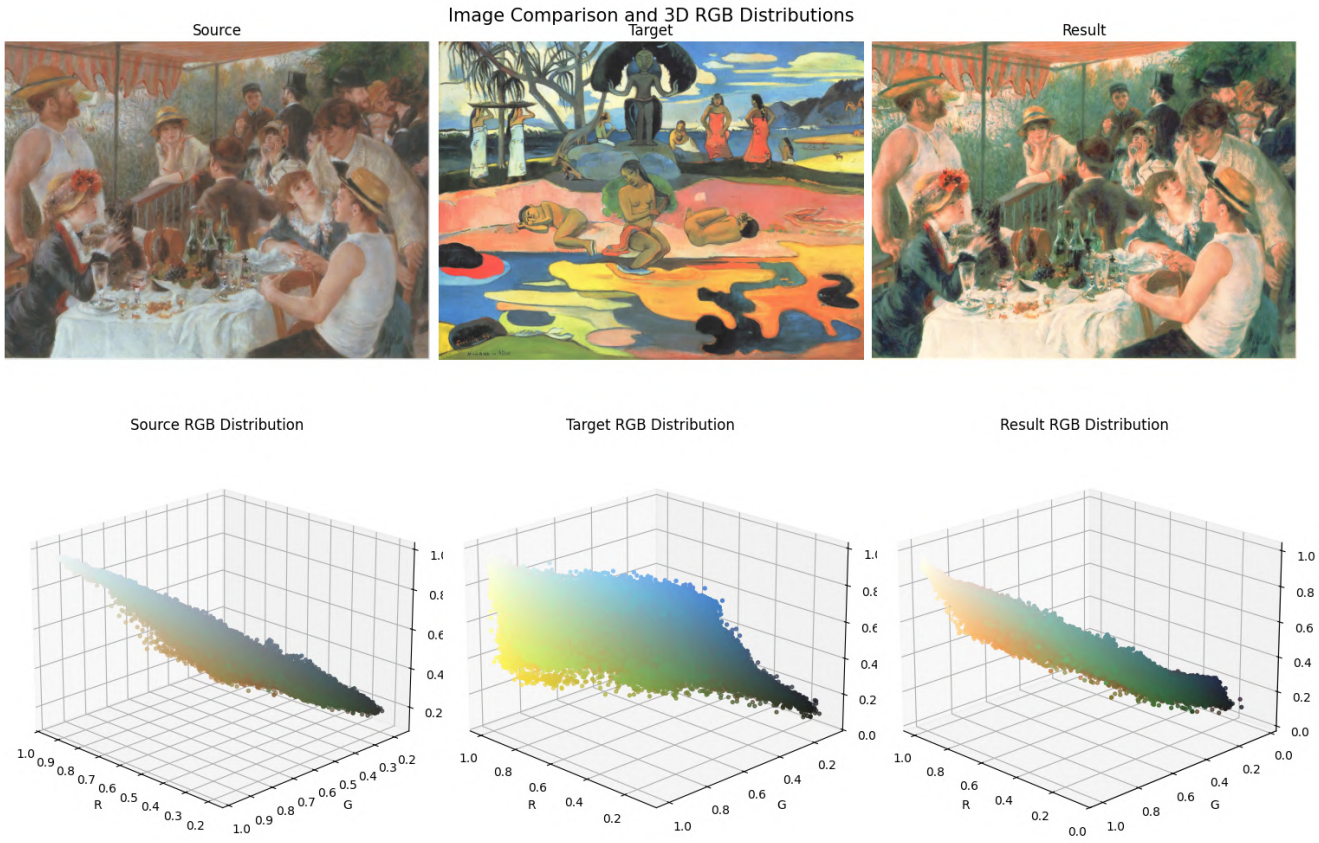


Figure 1: Histogram Matching (RGB Color Space).

To address this issue, we repeated the same procedure in the much less correlated CIE Lab color space. The result, shown in Figure 2, is significantly improved. Because Lab separates luminance from chromaticity and reduces inter-channel correlations, performing histogram matching independently on the L, a, and b components produces far more coherent color changes, and a closer resemblance to the target's overall color style.

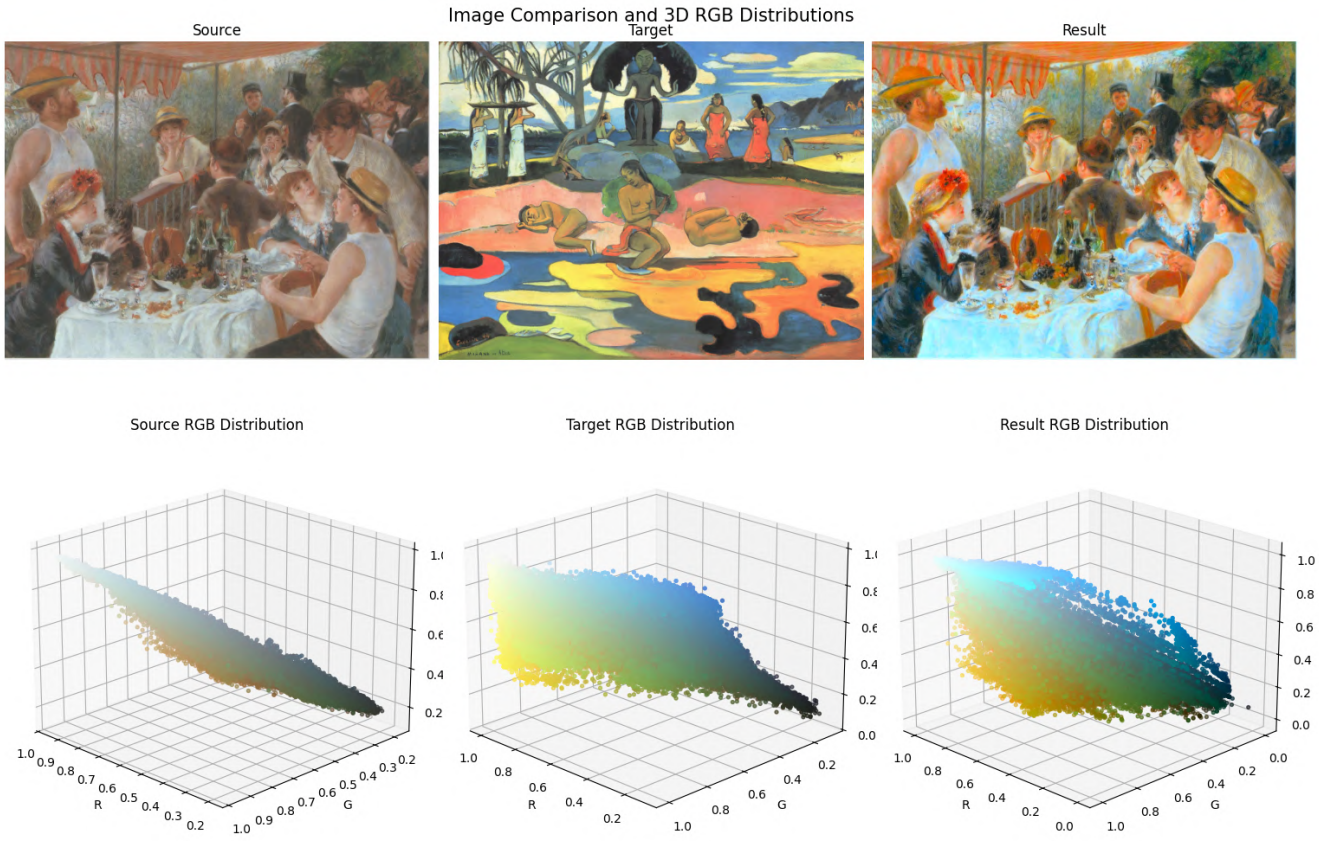


Figure 2: Histogram Matching (Lab Color Space).

Although histogram matching in Lab represents a clear improvement over RGB, the result is still limited in quality and does not reach the level achieved by the more sophisticated approaches presented in the following sections.

3 Mean and Covariance Matching

This approach was first introduced as a global statistical color transfer method by Reinhard et al. [1]. This method performs color transfer by aligning the first and second-order statistics (mean and covariance) of the source and target color distributions using an affine transformation in RGB color space. Colors are represented as vectors in $\mathbb{R}^3$, and the method aims to transform the source distribution to match the target's statistical profile.

Let X_s and X_t be the flattened color vectors of the source and target images, respectively. The mean vectors $\boldsymbol{\mu}_s$, $\boldsymbol{\mu}_t$ and covariance matrices Σ_s, Σ_t of these distributions are computed. To construct the affine transformation, we apply a whitening and coloring procedure:

1. **Whitening:** The source distribution is decorrelated and normalized by applying the inverse square root of its covariance matrix. This is done using eigenvalue decomposition:

$$\Sigma_s = U_s D_s U_s^T, \quad \Sigma_s^{-1/2} = U_s D_s^{-1/2} U_s^T$$

2. **Coloring:** The normalized source is then re-correlated to match the target distribution using the square root of Σ_t:

$$\Sigma_t = U_t D_t U_t^T, \quad \Sigma_t^{1/2} = U_t D_t^{1/2} U_t^T$$

3. **Affine Mapping:** The transformation matrix A is defined as:

$$A = \Sigma_t^{1/2} \Sigma_s^{-1/2}$$

The color vectors are then transformed as:

$$Y = A(X_s - \boldsymbol{\mu}_s) + \boldsymbol{\mu}_t$$

Figure 3 presents the result of the affine matching method. The output reflects an overall successful global style transfer.

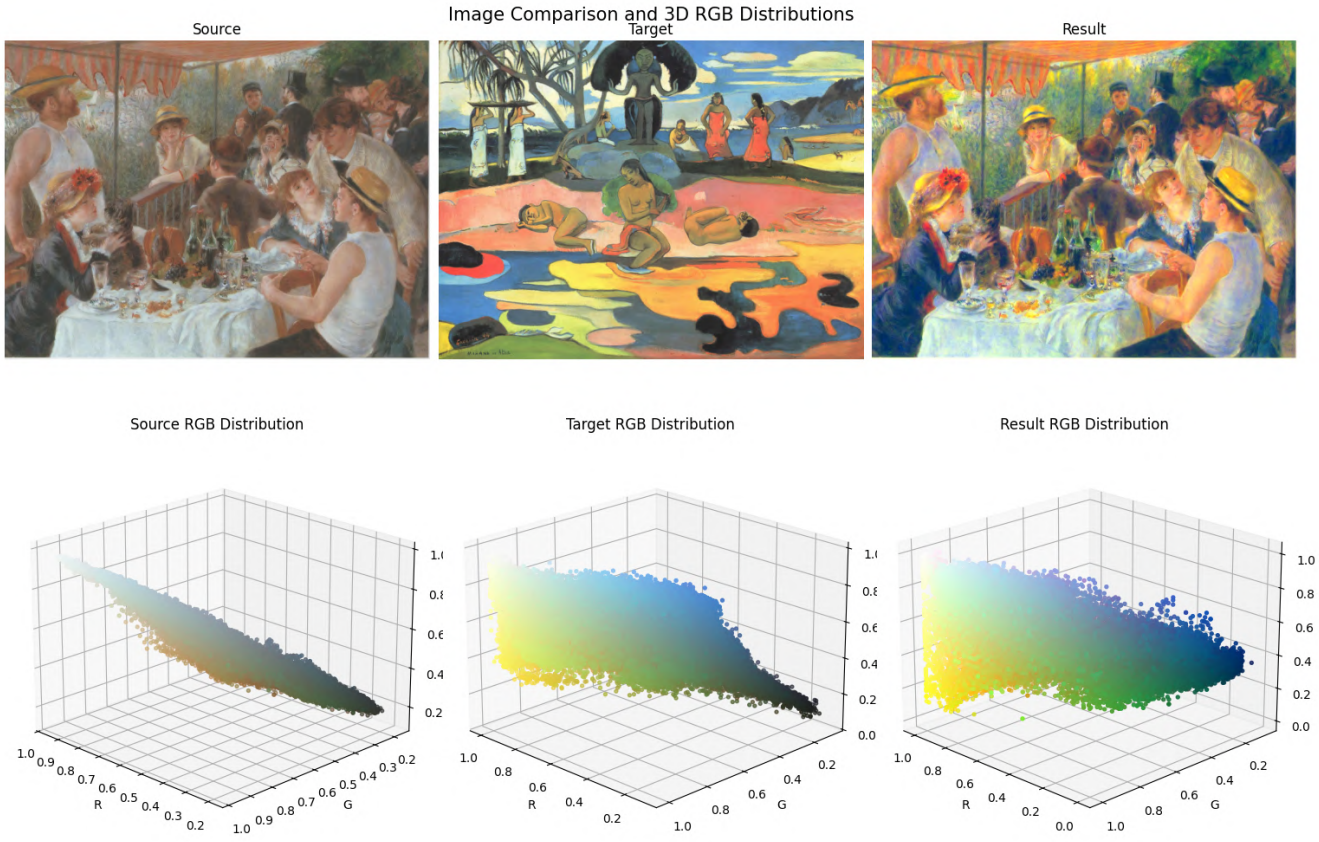


Figure 3: Mean and Covariance Matching Result.

Statistical Accuracy of the Affine Transform

To quantify how accurately the affine mean–covariance method matches the target color distribution, we evaluated the mean and covariance of the transformed image before and after the transformation, and compared it to the target image's statistics. The relative errors are computed as

$$\varepsilon_\mu = 100 \times \frac{\|\mu_{\text{current}} - \mu_t\|_2}{\|\mu_t\|_2}, \qquad \varepsilon_\Sigma = 100 \times \frac{\|\Sigma_{\text{current}} - \Sigma_t\|_F}{\|\Sigma_t\|_F}.$$

Source statistics:

$$\mu_s = (0.54146444,\ 0.48456544,\ 0.43353309)$$

$$\Sigma_s = \begin{pmatrix} 0.02297969 & 0.02092757 & 0.02024097 \\ 0.02092757 & 0.02486936 & 0.02614626 \\ 0.02024097 & 0.02614626 & 0.02888781 \end{pmatrix}$$

Target statistics:

$$\mu_t = (0.61045728,\ 0.55546412,\ 0.45481971)$$

$$\Sigma_t = \begin{pmatrix} 0.07374692 & 0.03952299 & 0.01011889 \\ 0.03952299 & 0.03872321 & 0.02267423 \\ 0.01011889 & 0.02267423 & 0.03144722 \end{pmatrix}$$

After affine transform:

$$\mu_{\text{aff}} = (0.61045728,\ 0.55546412,\ 0.45481971)$$

$$\Sigma_{\text{aff}} = \begin{pmatrix} 0.07374692 & 0.03952299 & 0.01011889 \\ 0.03952299 & 0.03872321 & 0.02267423 \\ 0.01011889 & 0.02267423 & 0.03144722 \end{pmatrix}$$

The affine transform matches the target almost perfectly:

$$\varepsilon_\mu = 2.0763 \times 10^{-10}\%, \qquad \varepsilon_\Sigma = 4.3 \times 10^{-13}\%.$$

Both the mean and covariance are numerically identical to the target up to floating-point precision. However, this evaluation implicitly assumes an unbounded color space: the affine transform is free to map colors anywhere in $\mathbb{R}^3$. In practice, the result must be clipped back to the displayable range (e.g. $[0, 1]$ in RGB). This clipping step necessarily perturbs the transformed distribution, as can be seen from the errors after clipping, which clearly increase despite the apparently perfect match above.

After clipping. Clipping introduces the expected distortion:

$$\mu = (0.60762077,\ 0.55524880,\ 0.45481972),$$

$$\Sigma = \begin{pmatrix} 0.07120816 & 0.03881159 & 0.01024040 \\ 0.03881159 & 0.03851934 & 0.02260442 \\ 0.01024040 & 0.02260442 & 0.03144719 \end{pmatrix}$$

$$\varepsilon_\mu = 0.30186386172319\%, \qquad \varepsilon_\Sigma = 2.47721683926154\%.$$

This suggests that some additional correction step is needed to compensate for clipping.

Critical Analysis

The Mean and Covariance Matching method is primarily valued for its **Computational Efficiency**, as it relies on simple statistical measures—the mean vector ($\boldsymbol{\mu}$) and the 3×3 covariance matrix (Σ)—to construct a fast, global affine transformation. This makes it an excellent choice for **real-time or fast batch processing**.

Conversely, its primary weakness is the **Unimodal Distribution Assumption**. Because the method only matches the first two moments, it fails when dealing with complex, **multimodal color palettes** (e.g., highly varied colors in a single image), leading to undesirable effects like **color leakage** or washed-out results.

4 Sliced Wasserstein Projection (SWP) Method

The Sliced Wasserstein Projection (SWP) method matches the full joint color distribution of the target image by repeatedly applying 1D optimal transport steps along random directions. In our implementation, SWP is formulated explicitly as a **gradient-descent** procedure on an empirical approximation of the sliced Wasserstein distance (SWD). At every iteration, we compute an estimated gradient of

the SWD using several random projections, and then update the source colors in the direction of the negative gradient.

The implementation and formulation of this method are based on the Numerical Tours notebook by Peyré and Régis [2].

Sliced Wasserstein Distance We use the sliced Wasserstein distance rather than other distances because most commonly used metrics for comparing distributions (such as the L^2 distance between histograms) completely ignore the geometric structure of the color distribution. They only compare frequencies or low-order statistics, and therefore fail to capture how colors are arranged in space. As a result, they cannot distinguish between two distributions that have identical histograms but very different shapes or modes in RGB space.

In contrast, the Wasserstein distance incorporates the underlying geometry of the data: it measures the minimal "effort" required to move the mass of one distribution onto the other. This makes it ideally suited for color transfer, where we want the entire cloud of colors of the source image to be transported toward the target distribution in a coherent, geometry-preserving way.

The full 3D Wasserstein distance is too expensive to compute on images, so we use its sliced variant (SWD), which reduces the problem to many 1D optimal transport problems.

Sliced Wasserstein Distance (SWD) Energy. Minimizing the SWD energy is the optimization procedure that makes the source distribution converge toward the target one. When the SWD cannot be reduced further, the algorithm has reached the closest distributional match achievable under the iterative SWP update rule.

The approximated SWD energy used in each iteration is

$$\widehat{E}_{\Theta}(f) = \frac{1}{d} \sum_{k=1}^{d} \left\| f\theta_k - q_k^{(l)} \right\|_2^2,$$

where $f\theta_k$ is the projection of f onto the direction θ_k, and $q_k^{(l)}$ is the 1D optimal transport–aligned projection of the target.

Although this formulation is expressed as a sum of 1D Wasserstein costs, it can be shown that minimizing this approximated energy is *equivalent* to the following update rule

$$f^{(\ell+1)} = (1-\tau)\, f^{(\ell)} + \tau\, \tilde{f}^{(\ell)} = f^{(\ell)} + \tau\, \Delta, \qquad \text{where } \Delta = \tilde{f}^{(\ell)} - f^{(\ell)}$$

and where $\tilde{f}^{(\ell)}$ is the vector obtained by matching $f^{(\ell)}$ to g along the sampled projection directions.

A full derivation of this equivalence, including the gradient computation and the simplified update rule is provided in Appendix C.

The core ideas of the method can be summarized as follows:

- **Random projections:** Each iteration samples d random unit directions $\{\theta_i\}$ in $\mathbb{R}^3$.
- **1D optimal transport:** For each θ_i, both the current source colors and the target colors are projected onto θ_i, sorted, and aligned using sorted percentile matching (the exact optimal transport map in 1D).

- **Gradient accumulation:** Each projection produces a displacement vector in the direction θ_i. These displacements are accumulated across all directions to form an estimated gradient of the SWD.
- **Gradient-descent update:** The color distribution is updated using

$$f \leftarrow f + \tau \cdot \frac{1}{d} \sum_{i=1}^{d} \Delta_i,$$

where τ is the step size controlling how aggressively the distribution moves at each iteration.

Over many iterations, this procedure reshapes the 3D color distribution of the image so that it approaches the target distribution, capturing high-order structure (such as multimodality), that cannot be reproduced by linear transforms.

To ensure stable convergence, the update magnitude is controlled by the step-size parameter τ. To study the influence of the step size τ on the stability and convergence speed of the SWP method, we performed a full sweep over the values $\tau \in \{0.1, 0.5, 1, 3, 5, 5.4, 5.8\}$ (with a max_iter parameter of 250 iterations). Figure 4 reports the evolution of the approximated SWD for each choice of τ.

The final SWD values and the stopping iteration for each τ are summarized below:

- $\tau = 0.1$ $\rightarrow$ Final SWD $= 0.401126$ after 250 iterations
- $\tau = 0.5$ $\rightarrow$ Final SWD $= 0.032846$ after 140 iterations
- $\tau = 1$ $\rightarrow$ Final SWD $= 0.021504$ after 85 iterations
- $\tau = 3$ $\rightarrow$ Final SWD $= 0.011509$ after 39 iterations
- $\tau = 5$ $\rightarrow$ Final SWD $= 0.007235$ after 29 iterations
- $\tau = 5.4$ $\rightarrow$ Final SWD $= 0.006043$ after 30 iterations
- $\tau = 5.8$ $\rightarrow$ Final SWD $= 0.007769$ after 74 iterations

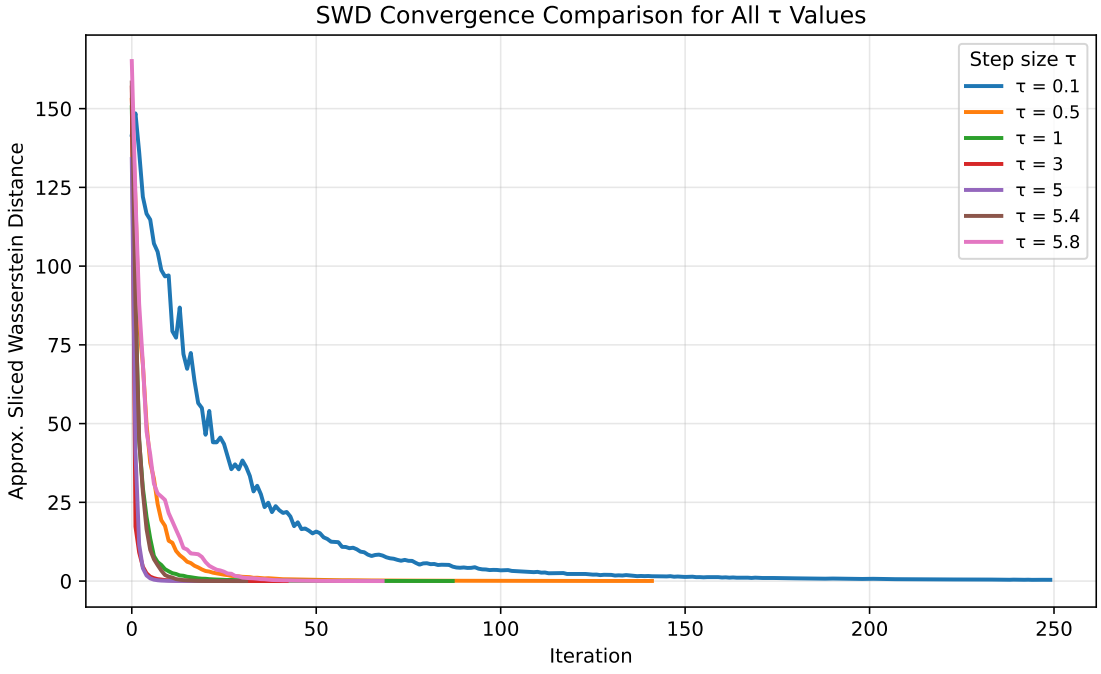


Figure 4: Convergence curves of the approximated SWD for different step sizes τ.

These experiments reveal a clear progression: small values of τ (such as 0.1) converge extremely slowly, while moderate values (0.5, 1, 3, 5) produce fast and smooth convergence. The step size $\tau = 5.4$ achieves the best overall behavior: it descends rapidly and reaches the lowest final SWD. In contrast, $\tau = 5.8$ starts to exhibit slight instability, and ultimately requires more iterations to converge than $\tau = 5.4$. This confirms that excessively large step sizes can degrade stability, whereas $\tau = 5.4$ provides the best balance between convergence speed and numerical stability. For this reason, all subsequent SWP experiments in this report use $\tau = 5.4$.

Convergence is monitored via a dual-epsilon scheme applied to the SWD curve. A short-horizon test checks the change $|\text{SWD}_t - \text{SWD}_{t-1}|$, while a long-horizon test evaluates $|\text{SWD}_t - \text{SWD}_{t-k}|$ over a wider window. Only when both tests remain below their thresholds for several consecutive iterations (after a minimum number of steps) is the algorithm considered converged. This prevents premature stopping while avoiding unnecessary computation.

Rewriting SWP in PyTorch provided massive speed improvements. The projection, sorting, displacement, and accumulation steps all exploit GPU parallelism, yielding speed-ups of roughly 60× over the NumPy prototype. This made it practical to use multiple directions per iteration and to experiment with larger images and parameter sweeps.

Figure 5 show the SWP result obtained. It produces strong stylistic alignment with the target, with excellent distribution matching visible in the RGB scatter plots. Note that for this method, it does not matter which color space we are working in (RGB or Lab), since the SWP method matches the entire color distribution of the image, no matter the working color space.

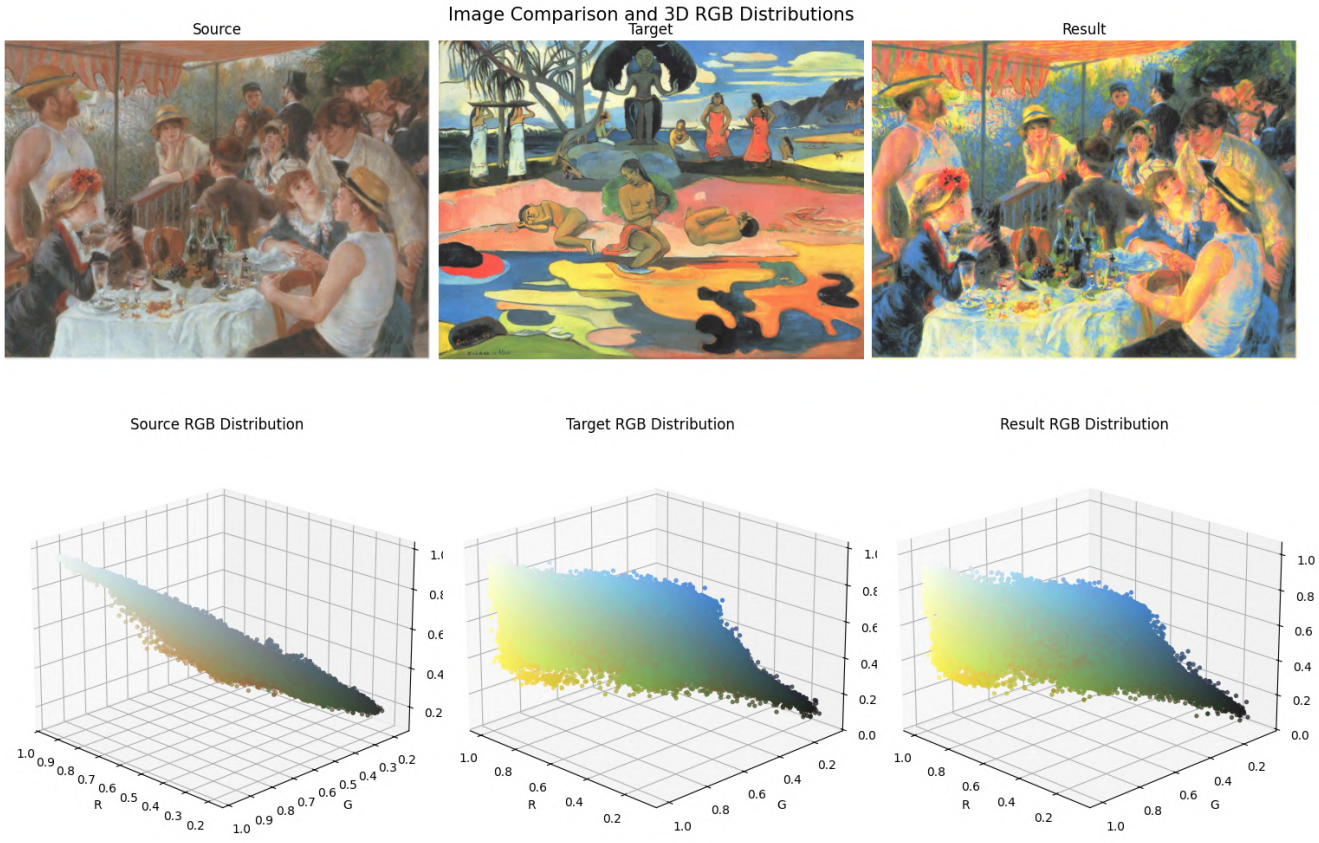


Figure 5: Sliced Wasserstein Projection method result.

Critical Analysis

The Sliced Wasserstein Projection (SWP) method offers the most expressive and accurate color transfer among the approaches studied, but it also comes with specific limitations and practical considerations.

- **Excellent Distribution Matching.** SWP is the only method that matches the *full joint color distribution* rather than only marginal histograms or low-order moments. As a result, it reproduces complex color palettes, multimodal distributions, and subtle stylistic variations that linear or per-channel methods cannot capture.
- **Color-Space Independence.** Because SWP operates directly on the empirical RGB (or Lab) distribution and matches 1D projections, its performance is largely independent of the chosen working color space. Unlike histogram matching and affine methods, switching between RGB and Lab produces nearly identical results. This makes SWP more robust and predictable across different color representations.
- **High Computational Cost.** The main drawback is the computational complexity. Each iteration requires multiple sorts and projections, and convergence may require dozens or hundreds of iterations. A fast GPU implementation is essential; without PyTorch acceleration, SWP is impractical for high-resolution images.
- **Stochasticity and Stability.** Because each iteration uses random projection directions, the method is stochastic. This can cause minor fluctuations in the SWD curve, and it motivated the use of a dual-epsilon stopping rule to detect stable convergence and avoid premature termination.
- **Potential for Artifacts.** Although SWP accurately matches distributions, it may produce artifacts such as noise amplification or blotchiness, especially in regions of smooth color. These artifacts arise because SWP treats pixels independently and does not consider spatial structure. A post-processing method is therefore an important complement to SWP when cleaner outputs are needed. This will be discussed in more detail later on in this report.

In summary, SWP provides the most faithful and flexible color distribution matching, significantly outperforming histogram matching and affine statistical alignment in terms of accuracy and expressiveness. However, this comes at the cost of higher computation and the risk of spatial artifacts, which motivates the use of regularization methods.

5 Handling Images of Different Sizes

The source and target images often have different numbers of pixels. Since the methods operate on pixel distributions rather than pixel positions, we handle size differences differently depending on the method.

5.1 Histogram Matching and SWP

Both histogram matching and SWP rely on matching quantiles. When the source has N pixels and the target has M pixels, we:

- sort the projected/source values (N samples) and the target values (M samples),
- build normalized quantile grids p_f and p_g,
- interpolate the target's sorted values at the source's quantile positions:

$$g_{\text{interp}} = \text{interp}(p_f, p_g, g_{\text{sorted}}),$$

- use these interpolated values to construct the mapping (histogram matching) or displacement field (SWP).

This quantile interpolation naturally supports different image sizes without resampling or resizing.

5.2 Mean and Covariance Matching

The affine mean–covariance method does **not** require any special handling for different sizes, since it matches only global statistics (mean and 3×3 covariance). These are independent of the number of pixels, so the method works identically for images of any size.

6 Transportation Map Regularization for Artifact Reduction

Color transfer methods often introduce undesirable artifacts such as blotchy color regions and noise amplification. These artifacts typically arise because the color transformation is applied independently to each pixel, without considering the spatial structure of the original image. The figure below illustrates the effect on our example.

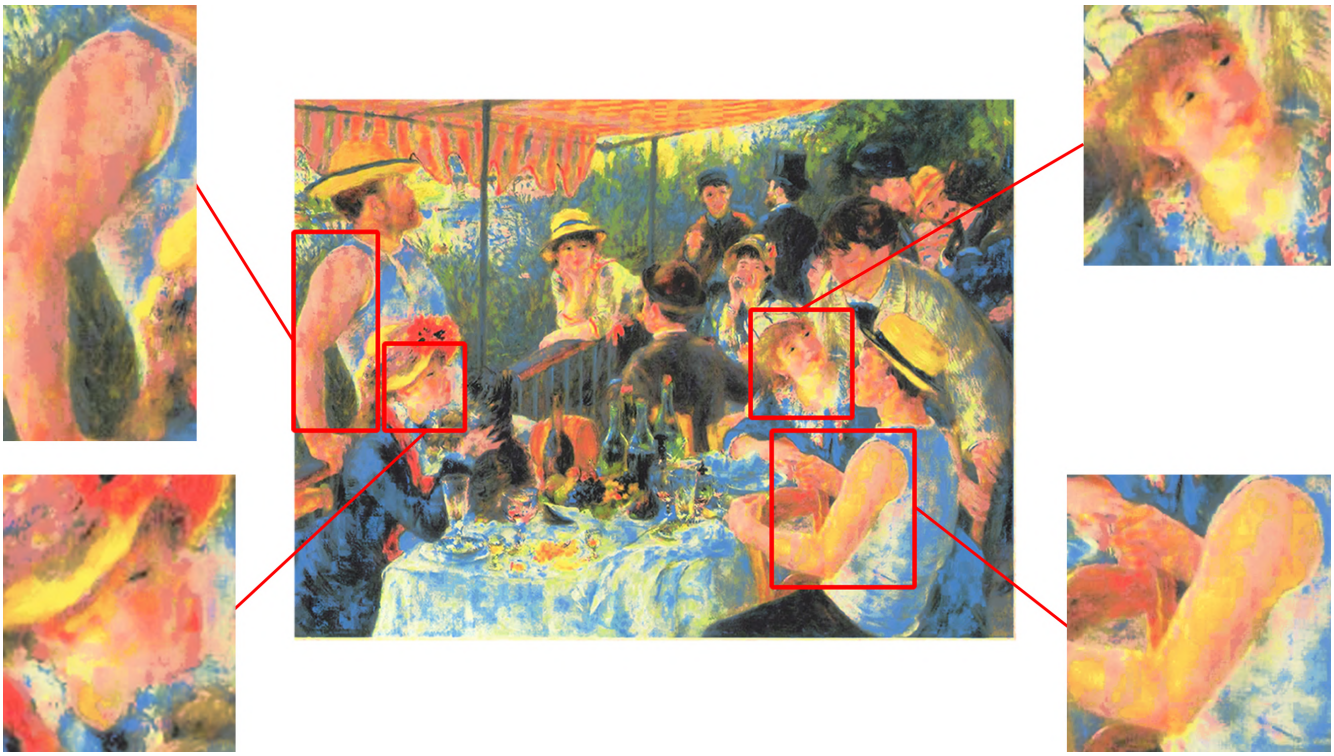


Figure 6: Artifacts produced by the Sliced Wasserstein method

To reduce these artifacts, we implemented the Transportation Map Regularization (TMR) method, which follows the formulation introduced by Rabin et al. [3]. TMR smooths the transportation map (the field of color changes produced by color transfer) in a way that preserves edges and local structures of the original image. Given u, the original image, and $T(u)$, the image after color transfer, we define the transportation map

$$\mathcal{M}(u) = T(u) - u.$$

This map represents how each pixel's color was shifted by the transfer method. The goal of TMR is to regularize this color-shift field, rather than smoothing the image directly. This preserves structural details while reducing color inconsistencies. This method applies an operator Y_u (the Yaroslavsky filter)

that performs a weighted average of the values in the transportation map, where the weights depend on similarities in the original image u. Formally:

$$Y_u(v)(x) = \frac{1}{C(x)} \sum_{y \in \mathcal{N}(x)} v(y)\, w_u(x, y),$$

with weights

$$w_u(x, y) = \exp\left(-\frac{\|u(x) - u(y)\|^2}{\sigma^2} \right).$$

This is equivalent to applying a joint bilateral filter where the guidance image is the original image u and the filtered image is the transportation map v. This ensures that neighboring pixels with similar color in u are strongly averaged, edges and color boundaries in u are preserved, and artifacts in the transportation map are reduced. In practice, we used OpenCV's jointBilateralFilter to implement this operator. A single application of Y_u removes some artifacts but not all. The paper shows that several iterations are beneficial. We define $v^{(0)} = \mathcal{M}(u)$ and iterate

$$v^{(k)} = Y_u\left(v^{(k-1)}\right).$$

The refined result after k iterations is

$$\mathrm{TMR}_u^{(k)}(T(u)) = u + v^{(k)}.$$

These iterations progressively reduce noise and inconsistencies while keeping edges aligned with the original image. The paper introduces a per-pixel convergence measure

$$C(x) = \|v^{(k)}(x) - v^{(k-1)}(x)\|.$$

A pixel stops updating when $C(x) < t$ with $t = 1$ (for n x 8-bit images). This means the color change between iterations is less than 1 level. In addition to this pixelwise rule, a global stopping condition is enforced by limiting the number of iterations to at most ten, so the procedure terminates either when all pixels have converged or when the iteration count reaches this maximum.

After applying this method on our example, the below result is obtained.

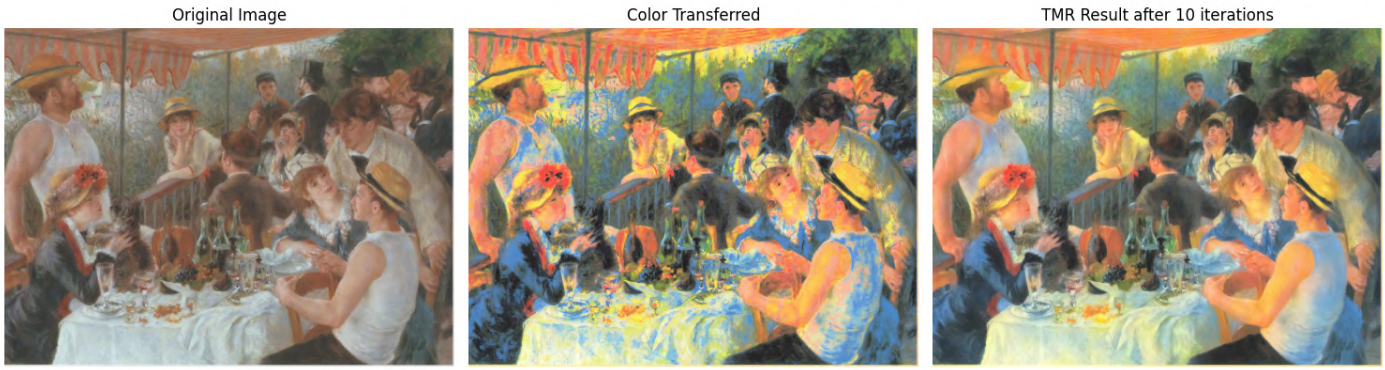


Figure 7: Result after applying 10 iterations of TMR

The artifacts are largely reduced, as demonstrated below.

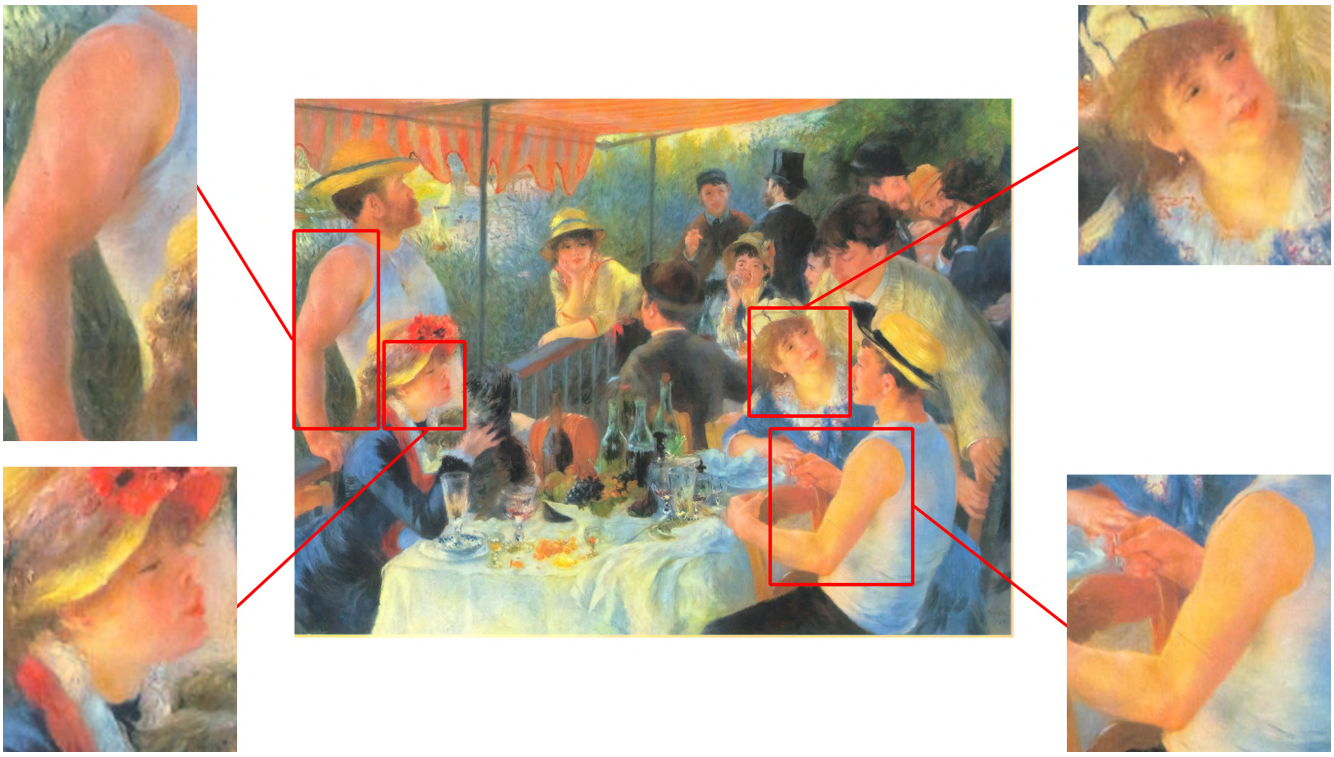


Figure 8: Close-ups showing artifact reduction after applying TMR

7 Limitations of Sliced Wasserstein Color Transfer

The sliced Wasserstein color transfer operates purely on global color statistics and applies the resulting mapping independently to each pixel. This design makes the method simple and flexible, but it also introduces several limitations.

The method can fail when the target image has a very different semantic structure and contrast level from the source. In the following example, a nearly monochrome, low-contrast portrait is matched to a brightly lit city scene with intense local contrasts and many distinct color modes (buildings, lights, reflections). The transfer pushes these saturated colors into all regions of the portrait, including the face and clothing, leading to unnatural tints and an overall overpainted look.

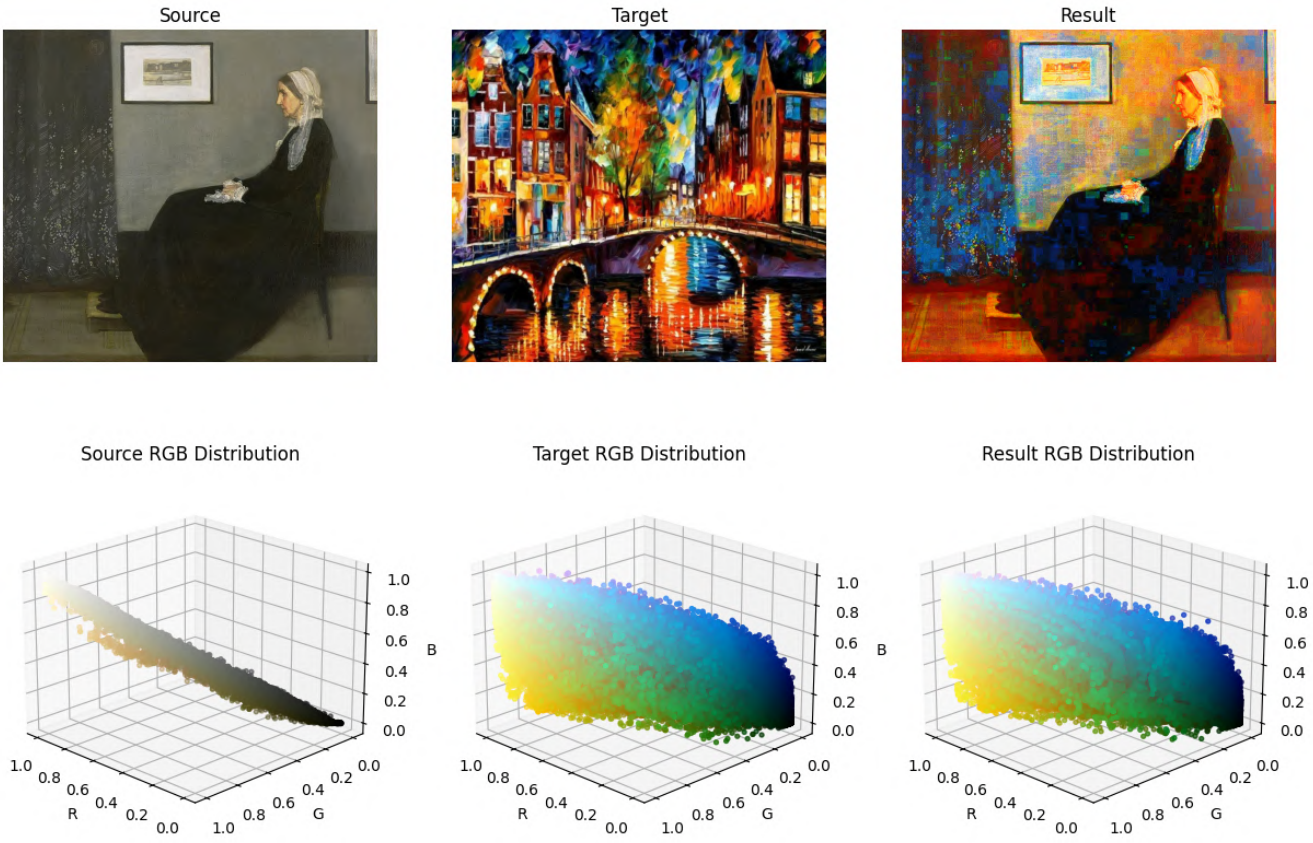


Figure 9: Bad Color Transfer Example 1

The method may also fail when the target assigns several very different colors to what is essentially the same object category. In the next example, the source trees are all naturally green, whereas the target fantasy landscape colors similar trees in a mixture of pink, purple, blue, and green. The sliced Wasserstein map has no constraint enforcing that all pixels belonging to the same tree in the source should receive similar colors; it simply matches pixel values. As a result, a single tree in the result can break into patches of incompatible hues, making the object look inconsistent even though the global color statistics closely follow the target.

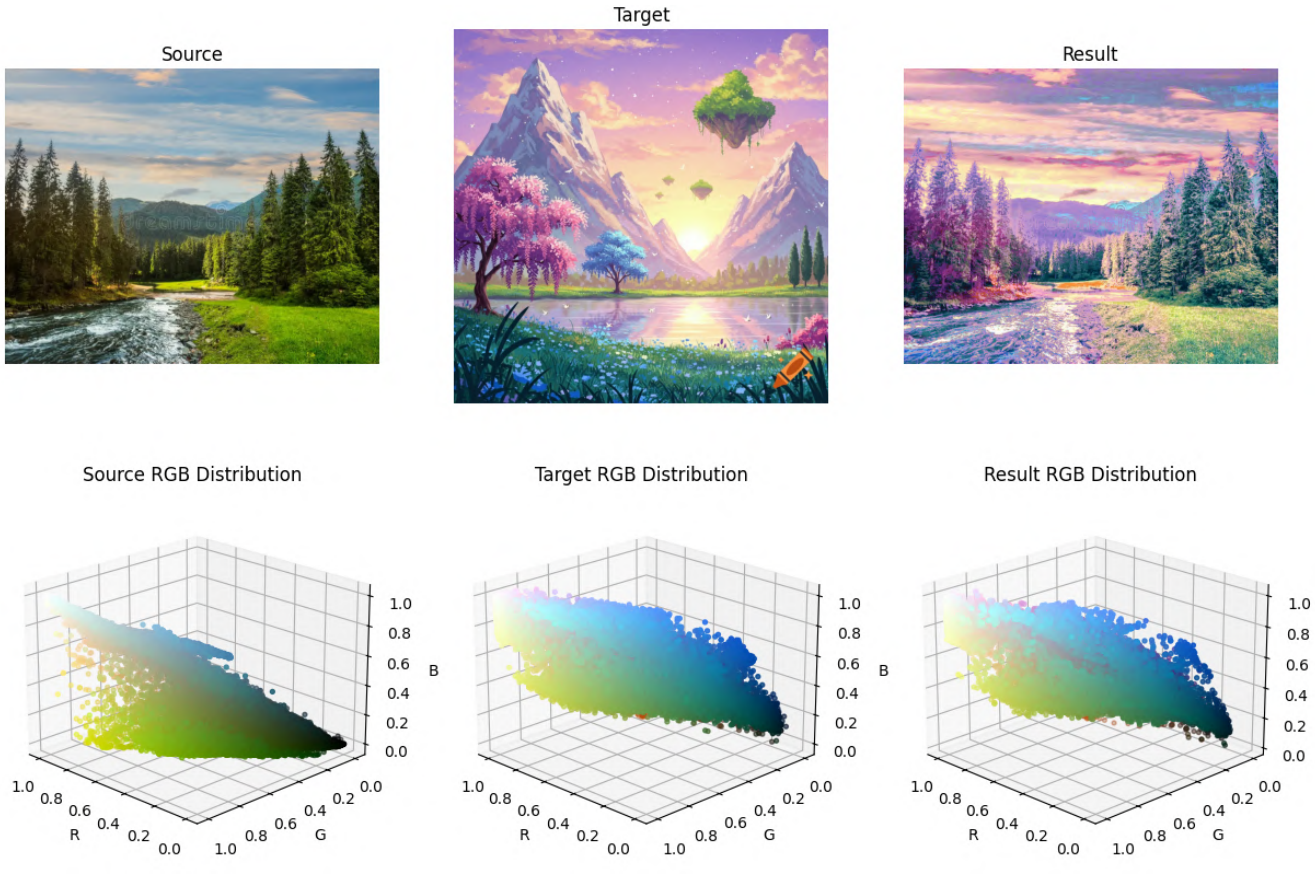


Figure 10: Bad Color Transfer Example 2

The sliced Wasserstein method can also break down when the target image has a color distribution that is highly uneven, full of clusters, gaps, and sharp jumps instead of smooth transitions. When the target's colors come from textured brushstrokes or stylized patterns, the distribution in RGB space becomes irregular, and the method must map smooth source gradients into this fragmented structure. Because the mapping is applied pixel-wise and has no way to preserve local coherence, smooth regions get pulled apart. This effect appears clearly in the example below, where the "Girl With a Pearl Earring" portrait's soft shading is forced into the jagged, brushstroke-driven color distribution of "The Starry Night", producing blocky patches and broken tonal transitions.

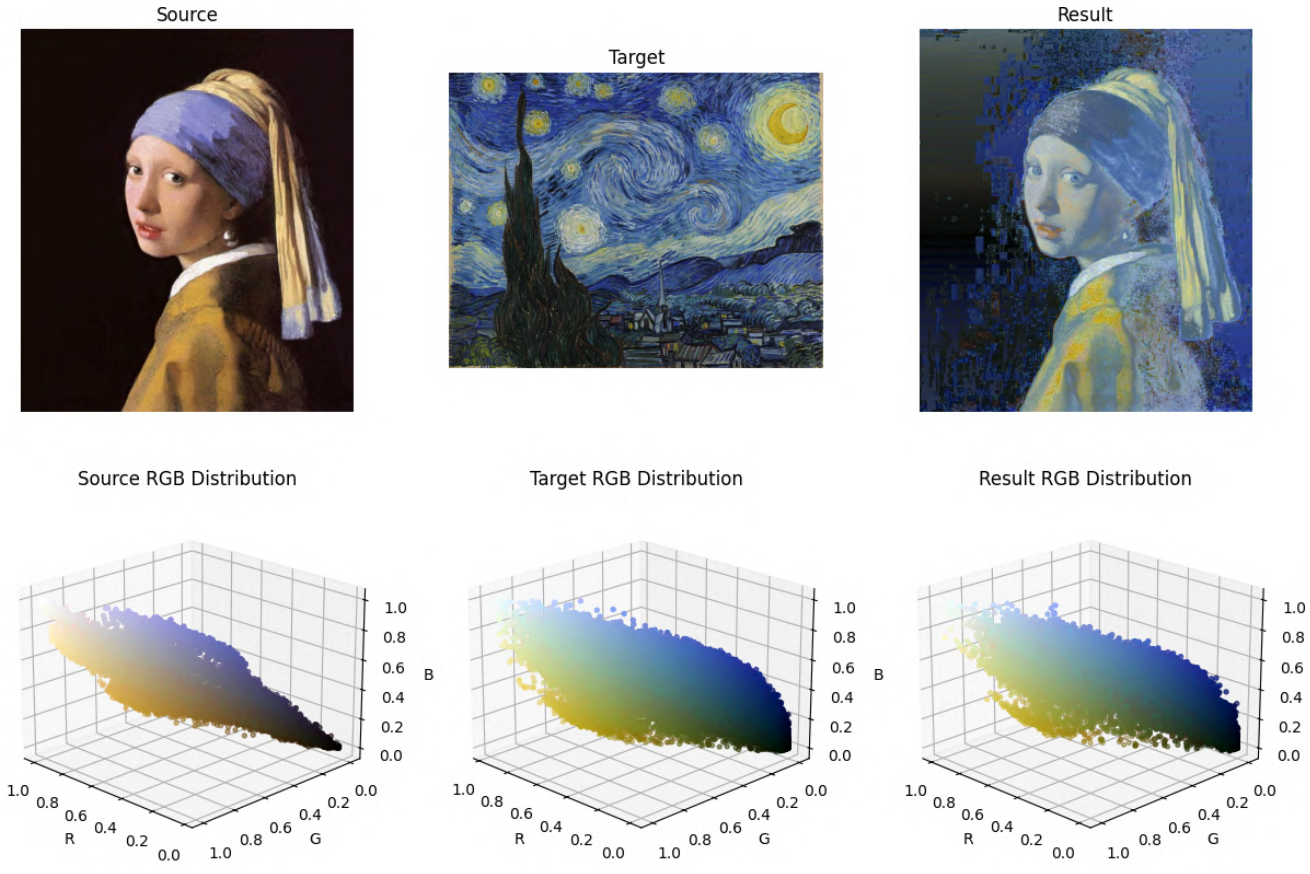


Figure 11: Bad Color Transfer Example 3

8 Conclusion

This report examined three increasingly sophisticated methods for transferring the color characteristics of a target image onto a source image: per-channel histogram matching, mean and covariance matching, and the Sliced Wasserstein Projection (SWP) method. Through visual comparisons and RGB distribution plots, we assessed the strengths and limitations of each approach.

Per-channel histogram matching, while simple to implement, proved highly sensitive to the choice of working color space. When performed in RGB, where channels are strongly correlated, the method produced poor results with unnatural hues. Operating in Lab space yielded more coherent outcomes, yet the method remained limited by its inability to preserve inter-channel relationships or capture the overall structure of the target distribution.

Mean and covariance matching provided more stable and natural color transfer. By aligning the first two statistical moments through a whitening–coloring transform combined with iterative clipping, the method produced globally consistent recolorings in both RGB and Lab working spaces. However, its reliance on only mean and covariance left it incapable of reproducing richer or multimodal color palettes present in the target image.

The SWP method achieved the closest match to the target's full color distribution. By iteratively aligning 1D projections using optimal transport and updating the color distribution via gradient-descent steps, SWP produced results that aligned nearly perfectly with the target's RGB distribution. Its performance was consistent across RGB and Lab spaces and benefitted greatly from a GPU-accelerated PyTorch implementation. Nevertheless, SWP introduced noticeable artifacts such as blotchiness and

noise amplification, especially in smooth regions, underscoring its lack of spatial regularization.

To address these artifacts, we applied Transportation Map Regularization (TMR), a post-processing technique that smooths the transportation map rather than the image itself. Using a joint bilateral filtering framework, TMR successfully reduced spatial inconsistencies while preserving edges and fine structures. The resulting images demonstrated visibly improved color coherence without sacrificing the stylistic characteristics imposed by the transfer.

Despite its efficiency, even the sliced Wasserstein approach remains constrained by its global, pixel-wise nature. Because the method matches distributions rather than semantic or spatial structures, it can introduce inconsistencies when the target palette differs strongly from the source in meaning, when identical objects in the scene are assigned multiple unrelated colors, or when the target's distribution is irregular due to stylized textures or abrupt tonal changes. These limitations underline that SWP captures global color statistics effectively but cannot guarantee perceptual coherence.

Appendix A

Additional SWP Color Transfer Examples

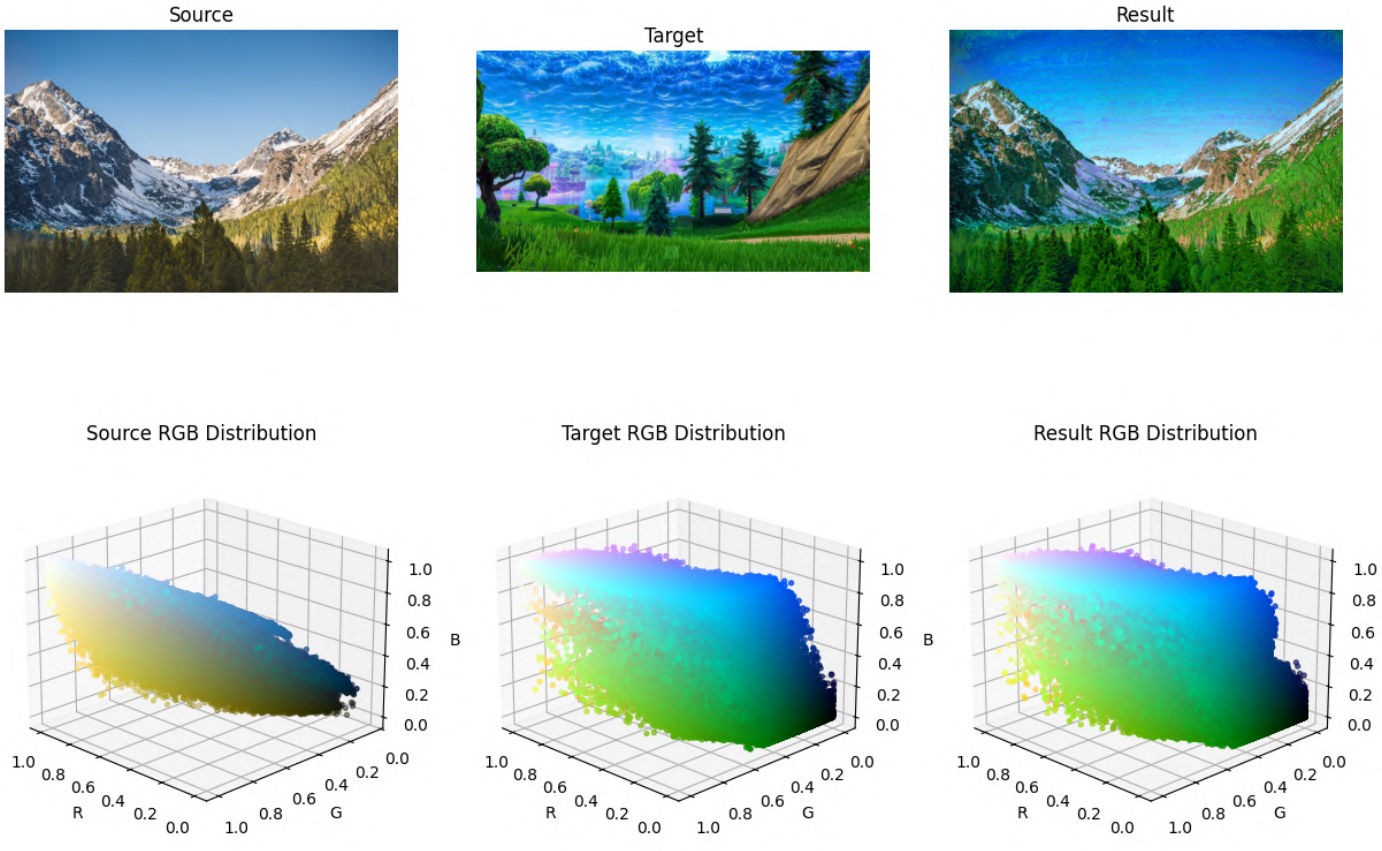


Figure A.1: Color Transfer Example 1

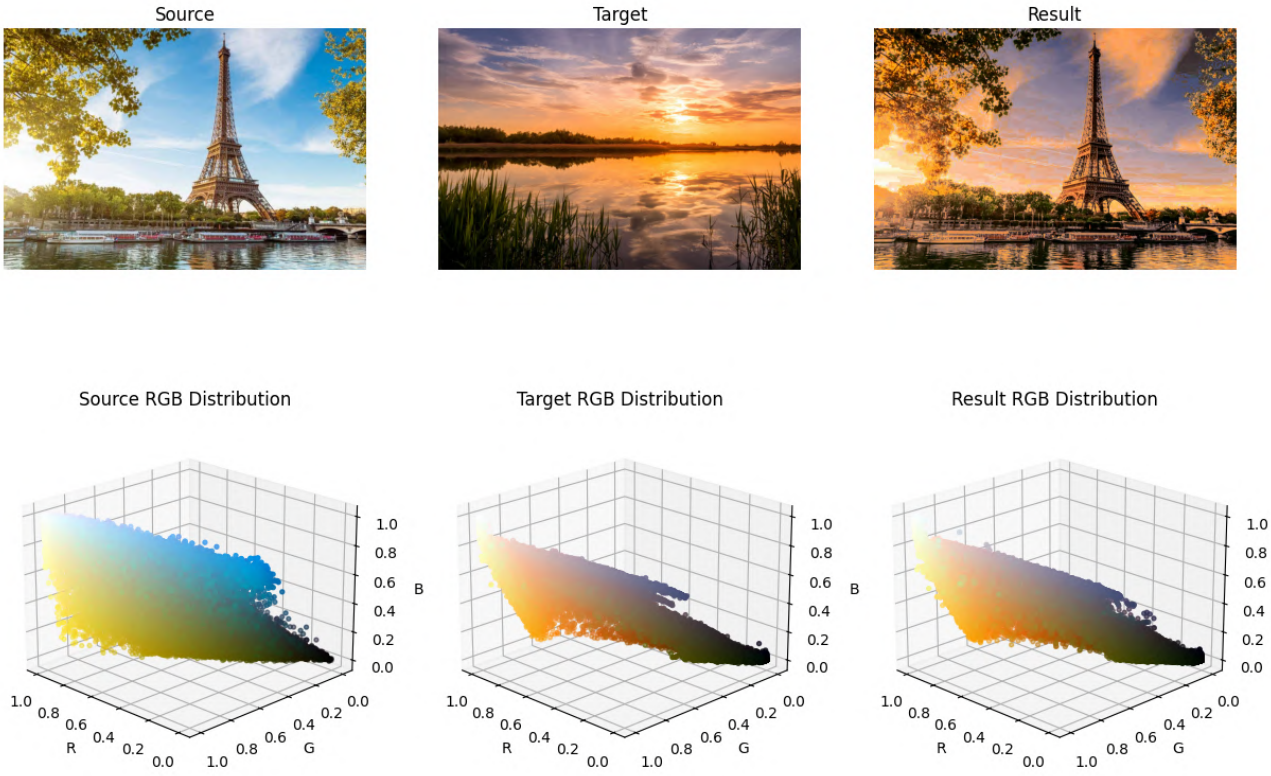


Figure A.2: Color Transfer Example 2

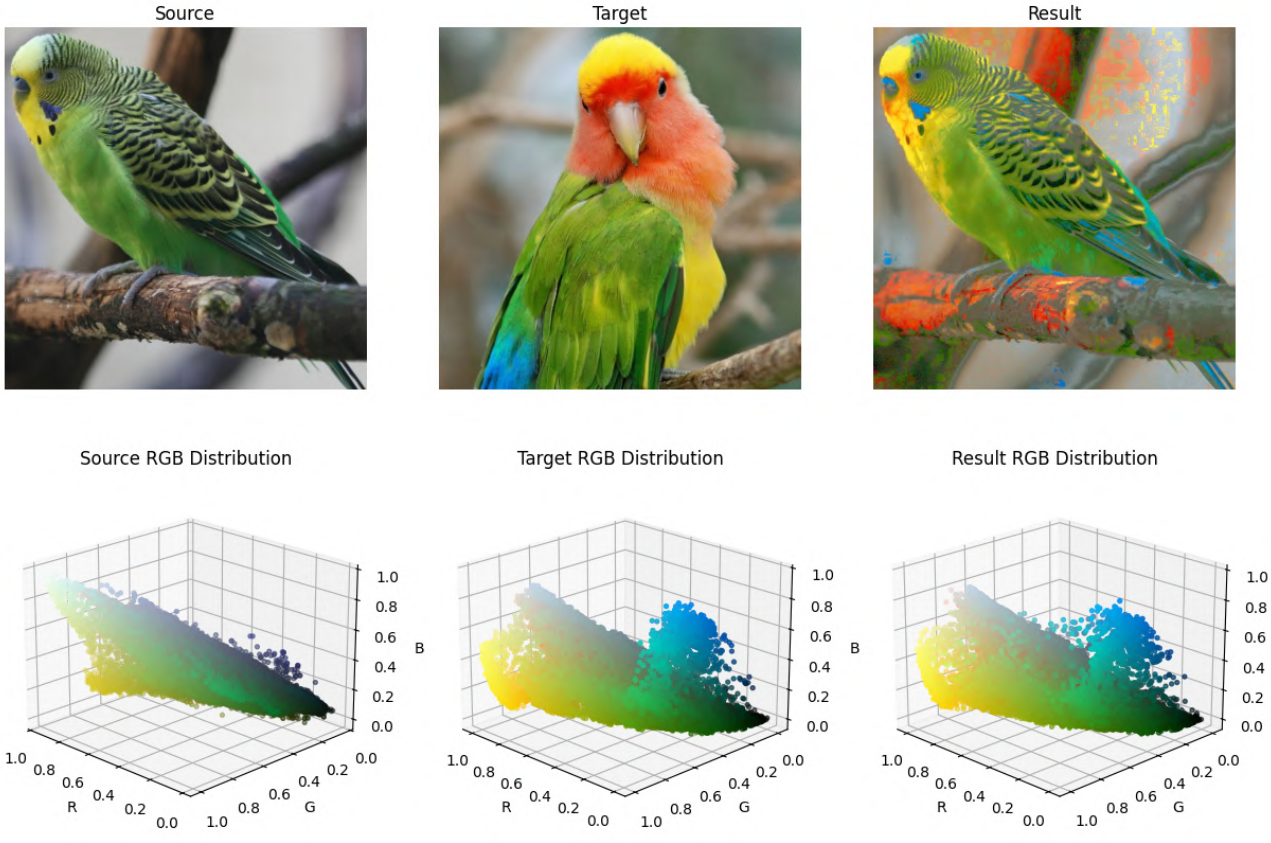


Figure A.3: Color Transfer Example 3

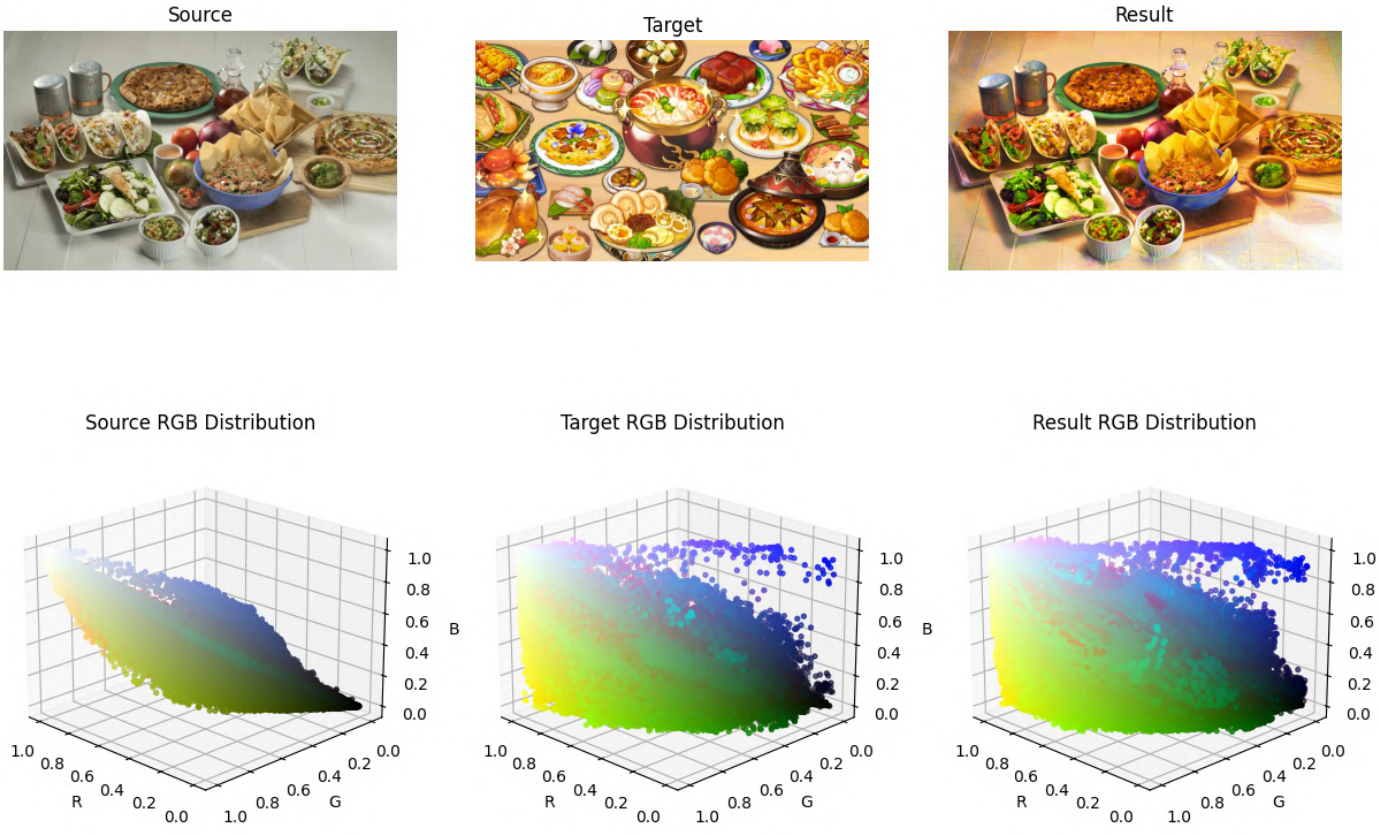


Figure A.4: Color Transfer Example 4

Appendix B

TMR Applied to Previous Examples



Figure B.1: TMR Result on Example 1



Figure B.2: TMR Result on Example 2



Figure B.3: TMR Result on Example 3



Figure B.4: TMR Result on Example 4

Appendix C

Derivation of the SWD Gradient

First, we flatten f and g into matrices of size $n \times 3$.

Matrix Convention

Symbol	Meaning	Shape
f	current image	$n \times 3$
g	target image	$n \times 3$
θ_k	projection direction (unit vector)	3×1
$\Theta = [\theta_1, \dots, \theta_d]$	all projection directions	$3 \times d$
$f\theta_k$	f projected on direction θ_k	$n \times 1$
$f\Theta$	f projected on all directions	$n \times d$

Projection and 1-D OT

For each projection k at iteration (l), compute the 1-D OT map:

$$\sigma^* : f^{(l)}\theta_k \longrightarrow g^{(l)}\theta_k.$$

Permute $g^{(l)}\theta_k$ using σ^* to obtain an aligned vector:

$$q_k^{(l)} = P\big(f^{(l)}\theta_k, g^{(l)}\theta_k\big) \in \mathbb{R}^{n\times 1}.$$

Stack all aligned projections:

$$Q^{(l)} = [q_1^{(l)}, \dots, q_d^{(l)}] \in \mathbb{R}^{n\times d}.$$

Approximated Wasserstein Energy

$$\hat{E}_\Theta(f) = \frac{1}{d}\sum_{k=1}^{d} \left\| f\theta_k - q_k^{(l)} \right\|_2^2 = \frac{1}{d} \left\| f\Theta - Q^{(l)} \right\|_F^2.$$

Gradient with Respect to f

$$\nabla_f \hat{E}_\Theta(f) = \frac{2}{d}\left(f\,\Theta - Q^{(l)}\right)\Theta^\top \in \mathbb{R}^{n\times 3}.$$

Define the reconstructed inverse-projection image:

$$\tilde{f}^{(l)} = Q^{(l)}\Theta^\top.$$

Then:

$$\nabla_f \hat{E}_\Theta(f) = \frac{2}{d}\left(f\,\Theta\Theta^\top - \tilde{f}^{(l)}\right).$$

Now if $d = 3$ and the matrix Θ is composed of orthonormal vectors, this formula can be simplified to

$$\nabla_f \hat{E}_\Theta(f) = \frac{2}{d}\left(f - \tilde{f}^{(l)}\right).$$

since $\Theta^\top = \Theta^{-1}$ in this case.

Gradient Descent Update

Starting from $f^{(l)}$:

$$f^{(l+1)} = f^{(l)} - \tau \nabla_f \hat{E}_\Theta(f^{(l)}) = f^{(l)} - \frac{2\tau}{d}\left(f^{(l)} - \tilde{f}^{(l)}\right).$$

Let

$$\tau' = \frac{2\tau}{d}.$$

Final update rule:

$$f^{(l+1)} = (1 - \tau')\, f^{(l)} + \tau'\, \tilde{f}^{(l)}.$$

Bibliography

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[2] G. Peyré and L. Régis. "Optimal transport – matching & sliced wasserstein." Accessed: Nov. 21, 2025, Numerical Tours. [Online]. Available: https://github.com/gpeyre/numerical-tours/blob/master/matlab/optimaltransp_4_matching_sliced.ipynb.

[3] J. Rabin, J. Delon, and Y. Gousseau, "Removing artefacts from color and contrast modifications," *IEEE Transactions on Image Processing*, vol. 20, no. 11, pp. 3073–3085, 2011.